Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/12/2023
PI Name:	Costes, Sylvain Ph.D.		
Project Title:	Blood-based Multi-scale Model for Cancer Risk from GCR in Genetically Diverse Populations		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sylvain.costes@bmsis.org	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	650-604-5343
Organization Name:	Blue Marble Space Institute of Science		
PI Address 1:	600 1st Avenue, 1st Floor		
PI Address 2:			
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98104	**Congressional District:**	7
Comments:	NOTE: After retiring from NASA, the PI joined Blue Marble. Dr. Costes was previously at Lawrence Berkeley National Laboratory until December 2016.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	02/04/2016	**End Date:**	10/01/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2022 per PI/CoI information (Ed., 2/4/22) NOTE: End date changed to 10/1/2021 per Space Radiation (Ed., 8/2/21) NOTE: End date changed to 9/30/2021 per L.Lewis/ARC HRP (Ed., 12/9/20) NOTE: Extended to 5/31/2021 per L. Lewis/ARC HRP (Ed., 9/24/20) NOTE: Extended to 5/31/2020 per PI (Ed., 11/15/19) NOTE: Extended to 9/30/2019 per F. Hernandez/ARC (Ed., 2/18/19)		

Key Personnel Changes/Previous PI:	NOTE (January 2018): The lab moved from Lawrence Berkeley National Lab (LBNL) to NASA Ames Research Center in 2017, where it was established as the Radiation Biophysics Lab in Space Biosciences Division. Dr. Costes will continue collaborating with LBNL and some funding will be left at LBNL to cover more plate processing in collaboration with Dr. Weil, Colorado State University (CSU) and for support from Dr. Snijders for the writing of the animal data. April 2017 report: - Elodie Guiet was a full time technician with a Bachelor in microbiology and biotechnology, working on this project from March 2016 until February 2017 -- she did not stay on the project when the lab moved to NASA Ames ; - Louise Viger was a Postdoc working partly on this project from June 2016 to January 2017 -- she was only here for a quick postdoc, focused primarily on modeling ; - Charlotte Degorre was a Postdoc who helped executing BNL run 16C -- visiting scientist for 1 month ; - Sebastien Penninckx was a PhD student who has been helping on data analysis -- visiting scientist for 3 months ; - Shayoni Ray is a new recruit at NASA Ames, postdoctoral fellow working on doing genomic analysis between animal DNA repair phenotypic data and their individual genes -- new postdoc full time at NASA Ames, started on April 10 2017 - Left in 2019 - Eloise Pariset was on the project until January 2020
COI Name (Institution):	
Grant/Contract No.:	Internal Project--ARC ; NNJ16HP24I
Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE (January 2018): The lab moved from Lawrence Berkeley National Lab (LBNL) to NASA Ames Research Center in 2017, where it was established as the Radiation Biophysics Lab in Space Biosciences Division. Dr. Costes will continue collaborating with LBNL and some funding will be left at LBNL to cover more plate processing in collaboration with Dr. Weil, Colorado State University (CSU), and for support from Dr. Snijders for the writing of the animal data. Crews on future exploration missions to Mars and other destinations in our solar system will be exposed to acute low doses (<100 mSv) and chronic low doses (<0.1 mSv/min) of high-LET (linear energy transfer) ionizing radiation from solar particle events (SPE) and galactic cosmic radiation (GCR). Predicting cancer risk associated with these radiation types is a mission-critical challenge for NASA radiation health scientists and mission planners. Epidemiological methods lack sensitivity and power to provide detailed risk estimates for cancer, mainly because the number of exposed individuals to date is relatively small, limited to several hundred individuals exposed to trapped radiation in low Earth orbit and fewer than two dozen Apollo astronauts exposed to GCR for several days at a time. Moreover, population-based studies do not take individual radiation sensitivity into account, are sensitive to the presence of other confounding environmental insults, and require long follow-up times. In collaboration with the radiation Biodosimetry Laboratory and the modeling group at NASA Johnson Space Center and with the International Computer Science Institute (ICSI) at University of California (UC) Berkeley, our team will bring unique inter-disciplinary expertise to integrate the large array of cancer data generated over the past 25 years and archived by NASA under the various Human Research Program (HRP) funded projects. The main goal of this proposal is to identify factors influencing radiation-induced carcinogenesis and integrate them into a multi-scale model already started at the Berkeley Lab that encompasses DNA damage response and inter-cellular signaling to predict cancer risk for any types of HZE (high energy particles). Because experimental data are dispersed across many different cancer models, radiation qualities, and measurement types, this project will also generate a complete set of experimental data designed to fully inform and validate the model. In this project, the model will impose the types of measurements being made, with a strong emphasis on well-established blood biomarkers. In our approach we hypothesize that genetic factors strongly influence risk of cancer from space radiation and that biomarkers reflecting DNA damage and inflammatory processes in the blood are great tools to predict risk and monitor potential health effects post-flight. By using blood as a surrogate organ, the proposed work will allow extrapolation of cancer risk from mice to humans. A cohort of 6 different strains of mice (collaborative cross-mouse) with expected sensitivity to ionizing radiation will be monitored for biomarkers and cancer after exposure to 0.3 Gy of 1 GeV/amu Fe particle and compared to 1 Gy exposure of gamma ray control. Because we favor larger number of animals per radiation condition, we selected only one dose and the most carcinogenic particle to prove the principle of our approach while validating our model on a complete set of ex-vivo data and in-vivo longitudinal data. The collaborative cross-mouse model used in this work was a resource from the low dose program at DOE (Department of Energy) developed by the Lawrence Berkeley National Laboratory that has made it possible for our team to examine the impact of genetic diversity in an animal model in a systematic and reproducible manner. In parallel, we propose to fully characterize the DNA damage response and cell death from ionizing radiation administered ex-vivo to 30 genetically different strains of mice and to 1000 human blood donors, matching the age and gender distribution of the astronaut population. Taken together, an array of ex-vivo phenotypic features will be associated to genetic traits across mice and humans as a function of age and gender. At the end of this proposal, our team will provide NASA with a model to estimate individualized risk for an astronaut before a flight as well as estimating the risk during the flight. Information generated in this proposal will also be useful to generate guidelines and suggest the best biomarkers to monitor the healthy recovery of astronauts post-flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A current radiobiology challenge is the ability to predict cancer risk associated with exposure to acute (<100 mSv) and chronic (<0.1 mSv/min) low doses of high-LET ionizing radiation. Epidemiological methods lack the sensitivity and power to provide detailed risk estimates for cancer, mainly because the astronaut cohort exposed to galactic cosmic rays (GCR) is relatively small. Moreover, population-based studies do not take individual radiation sensitivity into account, are affected by the presence of other confounding environmental insults, and require long follow-up times. We have hypothesized that characterizing the dose and time dependence of 53BP1 radiation induced foci (RIF) after exposure to a systematic array of X-ray doses and time points is sufficient to describe someone's ability to respond to any other LET. The main concept is that the non-physiological response to high doses of low-LET in cells can be used to predict the response to low doses of high-LET, and that the response to low and high doses of radiation is modulated by different pools of genes. Such work provides a new approach combining novel biomarkers with sophisticated mathematical analysis to better characterize individual sensitivity to space radiation. Once validated across mice and eventually a large cohort of humans, this approach could be generalized to establish individualized health risk management for astronauts and for the

	population at large being exposed to ionizing radiation.
Task Progress:	The main goal of this project was to identify the determinants of ex vivo sensitivity in humans and mice to DNA damage induced by simulated space radiation. We compared the radiosensitivity to multiple doses of high and low linear energy transfer (LET) ionizing radiation in ex vivo fibroblasts isolated from 15 Collaborative Cross and inbred mouse strains, followed by genome-wide association studies (GWAS) to identify the gene and pathway associations with initial DNA damage at 4 hours post irradiation, persistent DNA damage at 24+ hours post irradiation, the kinetics of DNA damage and repair, and baseline DNA damage without irradiation. We have also ranked the mouse strains based on radiosensitivity. We anticipate that our results will inform mouse strain selection for future radiobiology experiments and help identify the genes and pathways for biomarker and countermeasure development. We next analyzed human ex vivo immune cell sensitivity to multiple doses of high and low LET ionizing radiation in primary blood mononuclear cells isolated from 750+ healthy donors, followed by low coverage whole genome sequencing to identify the genes and pathways associated with initial DNA damage, persistent DNA damage, and baseline DNA damage. We have also analyzed the demographic associations with DNA damage and repair, and discovered an age-dependent increase in baseline DNA damage coupled with a reduction in ionizing radiation-induced DNA repair, as well as general negative association between baseline DNA damage and radiosensitivity. The project has resulted in numerous conference presentations as well as 5 primary research manuscripts (Penninckx et al. Radiat Res 2019; Pariset et al. Radiat Res 2020; Pariset et al. Cell Rep 2020; Penninckx et al. Radiat Res 2021; Cekanaviciute et al. Life Sci Space Res 2023) and contributed to 3 reviews (Ray et al. Int J Part Ther 2018; Nikitaki et al. Cancers 2020; Penninckx et al. NAR Cancer 2021) to date, with a manuscript on human genomic associations with radiosensitivity in preparation. The final presentation of the project was given at the NASA Human Research Program (HRP) Investigators' Workshop in January 2023. (Ed. Note: See Cumulative Bibliography for full citations).
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Articles in Peer-reviewed Journals	Cekanaviciute E, Tran D, Nguyen H, Lopez Macha A, Pariset E, Langley S, Babbi G, Malkani S, Penninckx S, Schisler JC, Nguyen T, Karpen GH, Costes SV. "Mouse genomic associations with in vitro sensitivity to simulated space radiation." Life Sci Space Res. 2023 Feb;36:47-58. https://doi.org/10.1016/j.lssr.2022.07.006 , Feb-2023
Articles in Peer-reviewed Journals	Scott RT, Sanders LM, Antonsen EL, Hastings JJA, Park S-m, Mackintosh G, Reynolds RJ, Hoarfrost AL, Sawyer A, Greene CS, Glicksberg BS, Theriot CA, Berrios DC, Miller J, Babdor J, Barker R, Baranzini SE, Beheshti A, Chalk S, Delgado-Aparicio GM, Haendel M, Hamid AA, Heller P, Jamieson D, Jarvis KJ, Kalantari J, Khezeli K, Komarova SV, Komorowski M, Kothiyal P, Mahabal A, Manor U, Garcia Martin H, Mason CE, Matar M, Mias GI, Myers JG, Nelson C, Oribello J, Parsons-Wingerter P, Prabhu RK, Qutub AA, Rask J, Saravia-Butler A, Saria S, Singh NK, Snyder M, Soboczenski F, Soman K, Van Valen D, Venkateswaran K, Warren L, Worthey L, Yang JH, Zitnik M, Costes SV. "Biomonitoring and precision health in deep space supported by artificial intelligence." Nat Mach Intell. 2023 Mar 23;5:196-207. Review. https://doi.org/10.1038/s42256-023-00617-5 , Mar-2023
Articles in Peer-reviewed Journals	Sanders LM, Scott RT, Yang JH, Qutub AA, Garcia Martin H, Berrios DC, Hastings JJA, Rask J, Mackintosh G, Hoarfrost AL, Chalk S, Kalantari J, Khezeli K, Antonsen EL, Babdor J, Barker R, Baranzini SE, Beheshti A, Delgado-Aparicio GM, Glicksberg BS, Greene CS, Haendel M, Hamid AA, Heller P, Jamieson D, Jarvis KJ, Komarova SV, Komorowski M, Kothiyal P, Mahabal A, Manor U, Mason CE, Matar M, Mias GI, Miller J, Myers JG, Nelson C, Oribello J, Park S-m, Parsons-Wingerter P, Prabhu RK, Reynolds RJ, Saravia-Butler A, Saria S, Sawyer A, Singh NK, Snyder M, Soboczenski F, Soman K, Theriot CA, Van Valen D, Venkateswaran K, Warren L, Worthey L, Zitnik M, Costes SV. "Biological research and self-driving labs in deep space supported by artificial intelligence." Nat Mach Intell. 2023 Mar 23;5:208-19. Review. https://doi.org/10.1038/s42256-023-00618-4 , Mar-2023
Articles in Peer-reviewed Journals	Wilson LJ, Kiffer FC, Berrios DC, Bryce-Atkinson A, Costes SV, Gevaert O, Matarèse BFE, Miller J, Mukherjee P, Peach K, Schofield PN, Slater LT, Langen B. "Machine intelligence for radiation science: Summary of the Radiation Research Society 67th annual meeting symposium." International Journal of Radiation Biology. 2023 Feb 6; https://doi.org/10.1080/09553002.2023.2173823 ; PMID 36735963 , Feb-2023
Articles in Peer-reviewed Journals	Ray S, Cekanaviciute E, Lima IP, Sorensen BS, Costes SV. "Comparing photon and charged particle therapy using DNA damage biomarkers." Int J Part Ther. 2018 Sep 21;5(1):15-24. https://doi.org/10.14338/IJPT-18-00018.1 , Sep-2018
Articles in Peer-reviewed Journals	Adamopoulos KI, Sanders LM, Hoarfrost A, Costes SV. "Deep cross-organism generalization of the physiological effects of spaceflight from mammalian model organisms to humans." Gravit Space Res. 2025 Jul 1;13(1):39-50. https://doi.org/10.2478/gsr-2025-0003 , Jul-2025